

INDU 211 · Worked Problems & Quantitative Analysis

Facilities Location & Transportation Quantitative Decision Guide

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1. The Mathematical Framework of Facility Location

In industrial and systems engineering, locating a facility (a manufacturing plant, regional distribution warehouse, or emergency medical center) requires rigorous mathematical modeling. Quantitative decision models balance two primary elements:

1. **Spatial Geometry (Distance Metrics):** How travel distance is measured between geographic coordinates (x_1, y_1) and (x_2, y_2) .
2. **Economic Optimization (Objective Function):** Minimizing the sum of volume-weighted transport costs, or minimizing the maximum response time.

Fundamental Equations

Euclidean Distance (Straight-line, L_2): $d_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Rectilinear Distance (Manhattan / Grid, L_1): $d_R = |x_1 - x_2| + |y_1 - y_2|$

Unweighted Center of Gravity: $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, $\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$

Weighted Center of Gravity: $\bar{x} = \frac{\sum_{i=1}^n Q_i x_i}{\sum_{i=1}^n Q_i}$, $\bar{y} = \frac{\sum_{i=1}^n Q_i y_i}{\sum_{i=1}^n Q_i}$

Rectilinear 1-Median Problem: $\min_{(x,y)} \sum_{i=1}^n Q_i (|x - x_i| + |y - y_i|)$

Transportation Model Objective: $\min Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$

2. Problem 1: Center of Gravity Analysis (Blood Bank Case Study)

Problem Statement (Lecture Slides 17–22)

A municipal public health authority needs to locate a new central blood bank to supply blood products to four regional hospitals: D_1 , D_2 , D_3 , and D_4 . The city street network follows an orthogonal Cartesian grid with the following hospital coordinates:

- **Hospital D_1 :** (2, 2)
- **Hospital D_2 :** (3, 5)
- **Hospital D_3 :** (5, 4)
- **Hospital D_4 :** (8, 5)



Questions:

1. **Scenario A (Equal Shipments):** Determine the optimal coordinates $(\bar{x}, \bar{y})$ if all four hospitals receive identical weekly blood shipments.
2. **Scenario B (Unequal Shipments):** Determine the optimal coordinates $(\bar{x}, \bar{y})$ if weekly delivery requirements vary as follows:
 - D_1 : 800 units/week
 - D_2 : 900 units/week
 - D_3 : 200 units/week
 - D_4 : 100 units/week
3. **Managerial Analysis:** Explain the physical and economic reason why the location shifted between Scenario A and Scenario B.

Step-by-Step Solution

Part 1: Scenario A (Equal Shipment Volumes)

When quantities are uniform ($Q_1 = Q_2 = Q_3 = Q_4$), the centroid formula is the simple arithmetic mean:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$$\bar{x} = \frac{2 + 3 + 5 + 8}{4} = \frac{18}{4} = \mathbf{4.50}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{y_1 + y_2 + y_3 + y_4}{4}$$

$$\bar{y} = \frac{2 + 5 + 4 + 5}{4} = \frac{16}{4} = 4.00$$

Result: The optimal unweighted blood bank location is $(\bar{x}, \bar{y}) = (4.50, 4.00)$.

Part 2: Scenario B (Varying Shipment Volumes)

When shipment quantities differ, we compute the volume-weighted centroid.

We construct a standardized calculation table:

Hospital (i)	x_i	y_i	Volume Q_i (units)	$Q_i \cdot x_i$	$Q_i \cdot y_i$
D_1	2	2	800	$800 \times 2 = 1,600$	$800 \times 2 = 1,600$
D_2	3	5	900	$900 \times 3 = 2,700$	$900 \times 5 = 4,500$
D_3	5	4	200	$200 \times 5 = 1,000$	$200 \times 4 = 800$
D_4	8	5	100	$100 \times 8 = 800$	$100 \times 5 = 500$
Total ($\sum$)	—	—	2,000	6,100	7,400

Now evaluate the weighted coordinates:

$$\bar{x} = \frac{\sum_{i=1}^4 Q_i x_i}{\sum_{i=1}^4 Q_i} = \frac{6,100}{2,000} = 3.05$$

$$\bar{y} = \frac{\sum_{i=1}^4 Q_i y_i}{\sum_{i=1}^4 Q_i} = \frac{7,400}{2,000} = 3.70$$

Result: The optimal weighted blood bank location is $(\bar{x}, \bar{y}) = (3.05, 3.70)$.

Part 3: Engineering & Managerial Interpretation

- In Scenario A, hospital D_4 at $(8, 5)$ pulled the unweighted centroid far to the east ($\bar{x} = 4.50$).
- In Scenario B, hospitals D_1 and D_2 account for **85% of total demand** ($800 + 900 = 1,700$ out of 2,000 units). Hospital D_4 represents only 5% of demand.
- Consequently, the weighted centroid moves significantly southwest from $(4.50, 4.00)$ to $(3.05, 3.70)$, positioning the facility close to the high-volume hospitals D_1 and D_2 . This drastically minimizes weekly delivery miles.

3. Problem 2: Rectilinear 1-Median Location Heuristic

Problem Statement (Textbook Example 4.3 / Slide 21)

Because city delivery takes place over an orthogonal grid, transportation costs are strictly proportional to **rectilinear distance**:

$$TC(x, y) = \sum_{i=1}^n Q_i (|x - x_i| + |y - y_i|)$$

1. Apply the **Rectilinear 1-Median Heuristic** to determine the optimal facility coordinates (x^*, y^*) for the weighted hospital problem ($Q_1 = 800, Q_2 = 900, Q_3 = 200, Q_4 = 100$).
2. Compare the total weekly transport distance incurred at the **Median solution** (x^*, y^*) versus the **Center of Gravity solution** $(3.05, 3.70)$.

Step-by-Step Solution

Part 1: Finding the Median Coordinates

Total weekly volume is $Q_{tot} = 2,000$ units. The median threshold is **50% of total volume**:

$$\text{Threshold} = \frac{1}{2} \sum_{i=1}^n Q_i = \frac{2,000}{2} = 1,000 \text{ units}$$

Step 1: Solve for x^* independently

Sort the destinations by ascending x -coordinate and accumulate weekly volume:

Destination	x_i	Volume Q_i	Cumulative Volume $\sum Q$	Condition ($\geq 1,000$)
D_1	2	800	800	No ($800 < 1,000$)
D_2	3	900	1,700	YES ($1,700 \geq 1,000$)
D_3	5	200	1,900	—
D_4	8	100	2,000	—

The cumulative volume crosses the 1,000-unit mark at $x = 3$.

$$\mathbf{x^* = 3}$$

Step 2: Solve for y^* independently

Sort the destinations by ascending y -coordinate and accumulate weekly volume:

Destination	y_i	Volume Q_i	Cumulative Volume $\sum Q$	Condition ($\geq 1,000$)
D_1	2	800	800	No ($800 < 1,000$)
D_3	4	200	1,000	YES ($1,000 \geq 1,000$)
D_2	5	900	1,900	—
D_4	5	100	2,000	—

The cumulative volume exactly reaches 1,000 at $y = 4$ and continues to $y = 5$. Any coordinate in the closed interval $[4, 5]$ achieves the theoretical minimum cost. Selecting $y^* = 5$ collocates the facility with Hospital D_2 (900 units), eliminating delivery distance for the single largest customer.

$$y^* = 5 \quad (\text{or any } y \in [4, 5])$$

Median Heuristic Solution: Locating at $(3, 5)$ (co-siting directly at Hospital D_2).

Part 2: Quantitative Distance Comparison

Evaluation at Median Solution $(3, 5)$:

$$TC(3, 5) = \sum_{i=1}^4 Q_i(|3 - x_i| + |5 - y_i|)$$

- To $D_1(2, 2)$: $800 \times (|3 - 2| + |5 - 2|) = 800 \times (1 + 3) = 800 \times 4 = 3,200$
- To $D_2(3, 5)$: $900 \times (|3 - 3| + |5 - 5|) = 900 \times (0 + 0) = 0$
- To $D_3(5, 4)$: $200 \times (|3 - 5| + |5 - 4|) = 200 \times (2 + 1) = 200 \times 3 = 600$
- To $D_4(8, 5)$: $100 \times (|3 - 8| + |5 - 5|) = 100 \times (5 + 0) = 100 \times 5 = 500$

$$TC(3, 5) = 3,200 + 0 + 600 + 500 = 4,300 \text{ unit-distance}$$

Evaluation at Center of Gravity Solution $(3.05, 3.70)$:

$$TC(3.05, 3.70) = \sum_{i=1}^4 Q_i(|3.05 - x_i| + |3.70 - y_i|)$$

- To $D_1(2, 2)$: $800 \times (|3.05 - 2| + |3.70 - 2|) = 800 \times (1.05 + 1.70) = 800 \times 2.75 = 2,200$
- To $D_2(3, 5)$: $900 \times (|3.05 - 3| + |3.70 - 5|) = 900 \times (0.05 + 1.30) = 900 \times 1.35 = 1,215$
- To $D_3(5, 4)$: $200 \times (|3.05 - 5| + |3.70 - 4|) = 200 \times (1.95 + 0.30) = 200 \times 2.25 = 450$
- To $D_4(8, 5)$: $100 \times (|3.05 - 8| + |3.70 - 5|) = 100 \times (4.95 + 1.30) = 100 \times 6.25 = 625$

$$\text{TC}(3.05, 3.70) = 2,200 + 1,215 + 450 + 625 = 4,490 \text{ unit-distance}$$

Key Industrial Engineering Finding:

The **Median location** (3, 5) yields 4,300 **unit-distances**, saving 190 **unit-distances per week** (a 4.2% cost reduction) compared to the Center of Gravity centroid (4, 490).

- **Theoretical Insight:** Center of Gravity minimizes *squared Euclidean distance* ($\sum d_i^2$). For rectilinear travel, the **1-Median method** is mathematically guaranteed to find the true cost-minimizing coordinate!

4. Problem 3: Multi-Facility Transportation Model (Plain View Manufacturing Co.)

Problem Statement (Lecture Slides 10–16)

Plain View Manufacturing Company manufactures computer peripheral units at existing factories in **Amarillo** and **Waco**, Texas. Demand at its three distribution warehouses in **San Antonio**, **Dallas**, and **Houston** has exceeded factory output. The president proposes constructing a new plant in **Huntsville**, Texas, bringing total system capacity to 2,000 units/month.

Operational Data:

- **Monthly Factory Capacities:**
 - Amarillo (F_1): 400 units
 - Waco (F_2): 1,000 units
 - Huntsville (F_3): 600 units
 - **Total Supply = 2,000 units**
- **Monthly Warehouse Demands:**
 - San Antonio (W_1): 300 units
 - Dallas (W_2): 900 units
 - Houston (W_3): 800 units
 - **Total Demand = 2,000 units** (*Supply equals Demand $\implies$ Balanced model*)

Unit Shipping Cost Matrix (c_{ij} in \$/unit):

Factory \ Warehouse	San Antonio (W_1)	Dallas (W_2)	Houston (W_3)	Factory Capacity (S_i)
Amarillo (F_1)	\$31	\$21	\$42	400
Waco (F_2)	\$20	\$21	\$30	1,000
Huntsville (F_3)	\$23	\$20	\$15	600
Warehouse Demand (D_j)	300	900	800	2,000

Questions:

1. Apply the **Least-Cost Assignment Method** to find a feasible initial shipment allocation and compute the total monthly transportation cost.
2. Demonstrate how an alternative tie-break choice affects the total cost.

Step-by-Step Solution

Iteration 1:

- Find the global minimum unit cost in the active matrix:

$$c_{33} = \$15 \quad (\text{Huntsville} \rightarrow \text{Houston})$$

- Maximum possible allocation:

$$x_{33} = \min(S_3, D_3) = \min(600, 800) = \mathbf{600 \text{ units}}$$

- Update balances:
 - Huntsville remaining capacity: $600 - 600 = 0$ (**Row 3 satisfied & eliminated**).
 - Houston remaining demand: $800 - 600 = 200$ units.

Iteration 2:

- Scan remaining active cells (Amarillo and Waco rows):
 - Amarillo: \$31, \$21, \$42
 - Waco: \$20, \$21, \$30
- Minimum active unit cost:

$$c_{21} = \$20 \quad (\text{Waco} \rightarrow \text{San Antonio})$$

- Maximum possible allocation:

$$x_{21} = \min(S_2, D_1) = \min(1,000, 300) = \mathbf{300 \text{ units}}$$

- Update balances:
 - San Antonio remaining demand: $300 - 300 = 0$ (**Column 1 satisfied & eliminated**).
 - Waco remaining capacity: $1,000 - 300 = 700$ units.

Iteration 3:

- Remaining active cells:
 - Amarillo $\rightarrow$ Dallas ($c_{12} = \$21$)
 - Amarillo $\rightarrow$ Houston ($c_{13} = \$42$)
 - Waco $\rightarrow$ Dallas ($c_{22} = \$21$)
 - Waco $\rightarrow$ Houston ($c_{23} = \$30$)
- Minimum cost is a tie at **\$21/unit**:
 - Option A: Allocate to Amarillo $\rightarrow$ Dallas ($c_{12} = \$21$).
- Maximum allocation:

$$x_{12} = \min(S_1, D_2) = \min(400, 900) = \mathbf{400 \text{ units}}$$

- Update balances:
 - Amarillo remaining capacity: $400 - 400 = 0$ (**Row 1 satisfied & eliminated**).
 - Dallas remaining demand: $900 - 400 = 500$ units.

Iteration 4:

- Only Waco remains active with 700 units of supply.
- Warehouse requirements remaining:
 - Dallas needs: 500 units ($c_{22} = \$21$)
 - Houston needs: 200 units ($c_{23} = \$30$)
- Complete assignments from Waco:

$$x_{22} = \mathbf{500 \text{ units}} \quad (\text{Dallas demand satisfied to 0})$$

$$x_{23} = \mathbf{200 \text{ units}} \quad (\text{Houston demand satisfied to 0})$$

- Waco capacity: $700 - 500 - 200 = 0$ (**Row 2 satisfied & exhausted**).
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Final Assignment Tableau (Primary Solution)

Factory \ Warehouse	San Antonio (W_1)	Dallas (W_2)	Houston (W_3)	Total Shipped
Amarillo (F_1)	—	400 [@ \$21]	—	400
Waco (F_2)	300 [@ \$20]	500 [@ \$21]	200 [@ \$30]	1,000
Huntsville (F_3)	—	—	600 [@ \$15]	600
Total Received	300	900	800	2,000

Total Monthly Shipping Cost:

$$\begin{aligned}TC &= (x_{33} \times c_{33}) + (x_{21} \times c_{21}) + (x_{12} \times c_{12}) + (x_{22} \times c_{22}) + (x_{23} \times c_{23}) \\&= (600 \times \$15) + (300 \times \$20) + (400 \times \$21) + (500 \times \$21) + (200 \times \$30) \\&= \$9,000 + \$6,000 + \$8,400 + \$10,500 + \$6,000 \\&= \mathbf{\$39,900} \text{ per month}\end{aligned}$$

(Note: In lecture slides, this is expressed as $6 \times 15 + 3 \times 20 + 4 \times 21 + 5 \times 21 + 2 \times 30 = \$399 \times 100 = \$39,900$).

Alternative Tie-Break Allocation (Lecture Slide 16)

In Iteration 3, suppose the tie at \$21 was broken by allocating all available capacity of Waco (700 units) to Dallas:

1. Allocate Waco $\rightarrow$ Dallas: $x_{22} = \min(700, 900) = \mathbf{700}$ units. (Waco exhausted).
2. Dallas still requires: $900 - 700 = 200$ units.
3. Allocate Amarillo $\rightarrow$ Dallas: $x_{12} = \min(400, 200) = \mathbf{200}$ units [@ \$21]. (Dallas satisfied).
4. Amarillo still has: $400 - 200 = 200$ units.
5. Houston still needs: $800 - 600 = 200$ units.
6. Force Amarillo $\rightarrow$ Houston: $x_{13} = \mathbf{200}$ units [@ \$42].

Total Cost for Alternative Solution:

$$\begin{aligned}TC_{\text{alt}} &= (600 \times \$15) + (300 \times \$20) + (200 \times \$21) + (700 \times \$21) + (200 \times \$42) \\&= \$9,000 + \$6,000 + \$4,200 + \$14,700 + \$8,400 \\&= \mathbf{\$42,300} \text{ per month}\end{aligned}$$

(Expressed as $\$423 \times 100$ in lecture slides).

Critical Managerial Takeaway:

Breaking the tie sub-optimally forced 200 units onto the very expensive Amarillo-to-Houston route at **\$42/unit**, incurring an extra **\$2,400/month (\$28,800/year)** in avoidable transportation waste! This underscores why

linear programming solvers (e.g., the Simplex-based Transportation Algorithm) are essential to find the true global optimum.

5. Problem 4: Quantitative Distance Metric Comparison & Grid Distortion

Problem Statement

A manufacturing engineer is analyzing part transit between Workstation A (10, 15) and Workstation B (70, 95) (coordinates in meters).

1. Calculate the straight-line Euclidean distance d_E .
2. Calculate the factory aisle Rectilinear distance d_R .
3. Compute the percentage elongation penalty imposed by the orthogonal factory grid.
4. If an AGV travels at 1.5 m/s , calculate the transit time difference between the two paths.

Step-by-Step Solution

Part 1: Coordinate Deltas

$$\Delta x = |70 - 10| = 60 \text{ meters}$$

$$\Delta y = |95 - 15| = 80 \text{ meters}$$

Part 2: Distance Calculations

- **Euclidean Distance:**

$$d_E = \sqrt{\Delta x^2 + \Delta y^2} = \sqrt{60^2 + 80^2} = \sqrt{3,600 + 6,400} = \sqrt{10,000} = \mathbf{100.0 \text{ meters}}$$

- **Rectilinear Distance:**

$$d_R = |\Delta x| + |\Delta y| = 60 + 80 = \mathbf{140.0 \text{ meters}}$$

Part 3: Elongation Penalty

$$\text{Penalty Ratio} = \frac{d_R}{d_E} = \frac{140}{100} = \mathbf{1.40}$$

$$\text{Percentage Increase} = \frac{140 - 100}{100} \times 100\% = \mathbf{+40.0\%}$$

Part 4: AGV Transit Time Difference

$$t_E = \frac{100 \text{ m}}{1.5 \text{ m/s}} = 66.67 \text{ seconds}$$

$$t_R = \frac{140 \text{ m}}{1.5 \text{ m/s}} = 93.33 \text{ seconds}$$

$$\Delta t = 93.33 - 66.67 = \mathbf{26.66} \text{ seconds per transit}$$

Industrial Engineering Impact: Over 500 AGV runs per day, rectilinear grid routing consumes an additional **3.7 hours of daily AGV travel time**, proving why aisle layout design directly affects fleet sizing and capital equipment expenditures.